JURONG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

CANDIDATE NAME	

CLASS	**18S**	EXAM INDEX	

CHEMISTRY

9729/04

Higher 2

Paper 4 Practical

16 August 2018
2 hours 30 minutes

Candidates answer on the Question paper.

Additional Materials: As listed in the Confidential Instructions

READ THESE INSTRUCTIONS FIRST

Write your name, class and exam index number on all the work you hand in.
Give details of the practical shift and laboratory where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use a HB pencil for any diagrams, graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Qualitative Analysis Notes are printed on pages 17 and 18.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	**13**
2	**16**
3	**12**
4	**14**
Total	**55**

This document consists of **18** printed pages and **0** blank page.

Answer **all** the questions in the spaces provided.

For Examiner's Use

1 Determination of the A_r of magnesium by a back titration method

FA 1 is an aqueous solution prepared by dissolving 0.215 g of magnesium ribbon in 30.0 cm^3 of 1.00 mol dm^{-3} hydrochloric and made up to 250 cm^3 with deionised water.

FA 2 is aqueous sodium carbonate containing 2.65 g dm^{-3} Na_2CO_3.

In this question, you will titrate **FA 2** with **FA 1** to determine how much hydrochloric acid was left over after the reaction with magnesium.

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

$$Na_2CO_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$$

Using the experimental results, you will determine the A_r of magnesium.

(a) Method

1. Fill the burette with **FA 1**.
2. Using a pipette, transfer 25.0 cm^3 of **FA 2** into the conical flask.
3. Add a few drops of methyl orange indicator.
4. Run **FA 1** from the burette into the conical flask until the yellow colour of the solution changes to orange.
5. Record your titration results in the space provided below. Make certain that your recorded results show the precision of your working.
6. Repeat points 1 to 5 as necessary until consistent results are obtained.

Results

[5]

(b) From your titrations, obtain a suitable volume of **FA 1** to be used in your calculations. Show clearly how you obtained this volume.

25.0 cm^3 of **FA 2** requires ………………. of **FA 1** for complete reaction. [1]

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(c) **Calculations**

(i) Calculate the amount of sodium carbonate in the 25.0 cm^3 of **FA 2** used in each titration. [A_r: Na, 23.0; C, 12.0; O, 16.0]

Amount of Na_2CO_3 in 25.0 cm^3 of **FA 2** = …………………. [1]

(ii) Use your answer in **(c)(i)** to calculate the amount of hydrochloric acid present in 250 cm^3 of **FA 1**.

[1]

Amount of HC*l* present in 250 cm^3 = ………………….

(iii) Calculate the amount of hydrochloric acid that reacted with the magnesium.

Amount of HC*l* that reacted with the magnesium = …………………. [1]

(iv) Hence calculate the relative atomic mass, A_r, of magnesium.

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A_r of magnesium = …………………. [1]

(d) The A_r of magnesium given in the Periodic Table is 24.3. Use this value given in the Periodic Table to calculate the percentage error of your A_r value of Mg obtained in **(c)(iv)**.

% error = …………………. [1]

(e) A solution of sodium hydroxide was prepared at the same concentration, in mol dm^{-3}, as **FA 2**. A student repeated the titration but replaced **FA 2** with this sodium hydroxide.

Explain the effect that replacing **FA 2** with this solution of sodium hydroxide would have on the volume of **FA 1** needed for the titration.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………………… [2]

[Total: 13]

2 Determination of enthalpy change of formation of magnesium oxide.

FA 3 is magnesium ribbon
FA 4 is magnesium oxide
FA 5 is 2.0 mol dm^{-3} hydrochloric acid

The enthalpy change of formation of magnesium oxide, ΔH_f(MgO), is the enthalpy change when one mole of magnesium oxide is formed from its elements in their standard states under standard conditions.

$$Mg(s) + \frac{1}{2} O_2(g) \rightarrow MgO(s) \qquad \Delta H_f(MgO)$$

To cause magnesium and oxygen to react together, they must be heated. The reaction then proceeds in an uncontrolled and highly exothermic fashion. It is not possible to measure this enthalpy change without the use of equipment such as a bomb calorimeter.

You are to perform an experiment by which you will determine this enthalpy change by an indirect route.

Both magnesium and magnesium oxide react exothermically with hydrochloric acid to form magnesium chloride. You will determine the molar enthalpy changes for **reaction 1** and **reaction 2**.

reaction 1 $\quad Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g) \qquad \Delta H_{rxn\ 1}$

reaction 2 $\quad MgO(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l) \qquad \Delta H_{rxn\ 2}$

You will then use Hess's Law to determine a value for the enthalpy change of formation of magnesium oxide, ΔH_f(MgO).

(a) Determining the enthalpy change of reaction 1 between FA 3 and FA 5.

Follow the instructions below to determine the maximum temperature change when a known mass of magnesium, **FA 3**, reacts completely with excess **FA 5**. Record all the mass readings in Table 2.1 and all measurements of time and temperature in the space provided below the table.

Method 1

1. Weigh the stoppered weighing bottle labelled **FA 3** and its contents. Record the total mass of weighing bottle and **FA 3**.
2. Use a 50 cm^3 measuring cylinder to transfer 50.0 cm^3 of **FA 5** into a Styrofoam cup. Place this cup inside a second Styrofoam cup which is held in a 250 cm^3 glass beaker to prevent it tipping over.
3. Stir the solution in the cup with the thermometer and measure its temperature. Record this temperature in your table in **(a)(i)** (this is the temperature at t = 0 min) and start the stop-watch. Repeat this measurement each minute for 3 minutes and record each temperature.
4. At the fourth minute, carefully add **FA 3** from the weighing bottle to **FA 5** in the cup. Quickly stir the mixture with the thermometer but **do not measure the temperature**.
5. Continue to stir the mixture. Measure and record the temperature at 4½ minutes, then **every half minute** until the temperature reaches a maximum, and then **every minute** until the tenth minute.
6. Empty and wash the Styrofoam cup. Rinse the cup with deionised water and dry it using a paper towel.
7. Reweigh the emptied weighing bottle and record its mass. Calculate and record the mass of **FA 3** used.

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(a) (i) Results

Record the required mass readings in Table 2.1 and in an appropriate format in the space below, record all measurements of time and temperature.

Table 2.1

Mass of weighing bottle with **FA 3** /g	
Mass of empty weighing bottle / g	
Mass of **FA 3** used / g	

Measurements of time and temperature:

[3]

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(a) (ii) Plot a graph of temperature (y-axis) against time (x-axis) on the grid below. You will use the graph to determine the theoretical temperature change at 4 minutes. The scale for temperature should extend at least 1 °C above your highest recorded temperature.

1. Draw a straight line of best-fit for the points before the fourth minute. Draw a second straight line of best-fit for the points after the fourth minute. Extrapolate both lines to the fourth minute.
2. From your graph, read the minimum and maximum temperature at the fourth minute. Record these values in the spaces below and determine a value for the maximum temperature rise, $\Delta \boldsymbol{T}$, at the fourth minute.

Temperature at 4^{th} minute

minimum = ………... ^oC

maximum = ….…….. ^oC

temperature rise, $\Delta \boldsymbol{T}$ = ……..….. K [4]

(iii) Use your $\Delta \boldsymbol{T}$ from **(a)(ii)** to calculate a value for the molar enthalpy change of **reaction 1** between **FA 3** and **FA 5**, $\Delta H_{rxn\ 1}$, in kJ mol^{-1}.
Assume that the specific heat capacity of the final solution is 3.75 J g^{-1} K^{-1} and that its density is 1.05 g cm^{-3}. [A_r: Mg, 24.3]

$\Delta H_{rxn\ 1}$ = …………………… kJ mol^{-1} [2]

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(b) Determining the enthalpy change of reaction 2 between FA 4 and FA 5.

A student carried out an experiment to determine the enthalpy change of **reaction 2**, $\Delta H_{rxn\ 2}$, using a known mass of **FA 4** that reacts with excess **FA 5**.

The student was provided with five pre-weighed samples of **FA 4**, labelled **v**, **w**, **x**, **y** and **z**, and carried out the experiment according to the following instructions.

Method 2

1. Use a 50 cm^3 measuring cylinder to transfer 50.0 cm^3 of **FA 5** into a Styrofoam cup. Place this cup inside a second Styrofoam cup which is held in a 250 cm^3 glass beaker to prevent it tipping over.
2. Stir the solution in the cup with the thermometer. Read and record its temperature. This is the initial temperature of **FA 4**, $\boldsymbol{T}_{initial}$.
3. Carefully add sample **v** to the **FA 5** in the cup.
4. Using the thermometer, stir the mixture continuously until it reaches its maximum temperature. Record the maximum temperature reached, $\boldsymbol{T}_{max}$.
5. Wash and carefully dry the Styrofoam cup.
6. Repeat steps 1 to 5 with the remaining samples **w**, **x**, **y** and **z**.

The student recorded all masses and temperatures in Table 2.2 below.

Table 2.2

	expt **v**	expt **w**	expt **x**	expt **y**	expt **z**
Mass of **FA 4** added to the cup, **m** / g	0.507	0.503	0.487	0.506	0.476
$\boldsymbol{T}_{initial}$ / °C	31.2	31.2	31.0	31.1	31.3
$\boldsymbol{T}_{max}$ / °C	38.7	38.6	38.2	38.2	38.3
Temperature change, $\Delta\boldsymbol{T}$ / °C					
$\dfrac{\Delta \boldsymbol{T}}{\mathbf{m}}$ / °C g^{-1}					

(i) Complete Table 2.2 by calculating values for $\Delta\boldsymbol{T}$ and $\Delta\boldsymbol{T}$/**m**. Leave values for $\Delta\boldsymbol{T}$/**m** to 3 significant figures.

(ii) Based on the calculated $\Delta\boldsymbol{T}$**/m** values in Table 2.2, determine an appropriate mean maximum temperature change, $\Delta\boldsymbol{T}_{ave}$, and mean mass of **FA 4**, $\boldsymbol{m}_{ave}$.

$\Delta\boldsymbol{T}_{ave}$ = …………………. °C $\boldsymbol{m}_{ave}$ = …………………… g [2]

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(iii) Use your results in **(b)(ii)** to calculate a value for the molar enthalpy change of **reaction 2** between **FA 4** and **FA 5**, $\Delta H_{rxn\ 2}$, in kJ mol^{-1}.
Assume that the specific heat capacity of the final solution is 3.75 J g^{-1} K^{-1} and that its density is 1.05 g cm^{-3}. [A_r: Mg, 24.3; O, 16.0]

$\Delta H_{rxn\ 2}$ = …………………… kJ mol^{-1} [2]

(c) Use your values for $\Delta H_{rxn\ 1}$ and $\Delta H_{rxn\ 2}$, together with the enthalpy change of formation of water, $\Delta H_f(H_2O)$, to calculate a value for the enthalpy change of formation of magnesium oxide, $\Delta H_f(MgO)$.

$$H_2(g) + \tfrac{1}{2} O_2(g) \rightarrow H_2O(l) \qquad \Delta H_f(H_2O) = -286 \text{ kJ mol}^{-1}$$

Note: If you were unable to calculate the enthalpy changes, assume that the value of $\Delta H_{rxn\ 1}$ is –457 kJ mol^{-1} and the value $\Delta H_{rxn\ 2}$ is –141 kJ mol^{-1}.

$\Delta H_f(MgO)$ = ………………… kJ mol^{-1} [2]

(d) Two methods, **Method 1** and **Method 2**, have been used to determine the maximum temperature change, $\Delta \boldsymbol{T}$, of an exothermic reaction. Which method would give a more accurate $\Delta \boldsymbol{T}$? Explain your answer.

……………………………………………………………………………………………...

………………………………………………………………………………………………

………………………………………………………………………………………………

[1]

[Total: 16]

For Examiner's Use

3 Planning

The solubility of calcium iodate(V), $Ca(IO_3)_2$, at 20 °C, is approximately 2.4 g dm^{-3}.

When solid calcium iodate(V) is added to water, a small amount dissolves to form a saturated solution, establishing an equilibrium between the undissolved salt and its aqueous ions.

$$Ca(IO_3)_2\,(s) \rightleftharpoons Ca^{2+}\,(aq) + 2IO_3^{-}\,(aq)$$

The equilibrium constant for the above solubility equilibrium, K_{sp}, is also known as the solubility product of calcium iodate.

$$K_{sp} = [Ca^{2+}(aq)]\,[IO_3^{-}(aq)]^2$$

This solubility product can be found by determining the equilibrium concentration of IO_3^- ions in a saturated solution of calcium iodate.

The exact concentration of IO_3^- ions is determined by titration. Excess aqueous KI and aqueous H^+ is first added to a sample of saturated calcium iodate solution to liberate iodine.

$$IO_3^-\,(aq) + 5I^-\,(aq) + 6H^+\,(aq) \rightarrow 3I_2\,(aq) + 3H_2O\,(l)$$

The iodine liberated in the resulting mixture is then titrated with sodium thiosulfate, $Na_2S_2O_3$ of known concentration.

$$I_2\,(aq) + 2S_2O_3^{2-}\,(aq) \rightarrow 2I^-\,(aq) + S_4O_6^{2-}\,(aq)$$

(a) Using the information given above, you are required to write a plan to determine the solubility product, K_{sp}, of calcium iodate, $Ca(IO_3)_2$, at 20 °C.

You may assume that you are provided with:

- solid sodium thiosulfate crystals, $Na_2S_2O_3.5H_2O$ (M_r = 248.2)
- solid calcium iodate, $Ca(IO_3)_2$
- aqueous potassium iodide, KI, of about 0.2 mol dm^{-3}
- aqueous hydrochloric acid, HC*l*, of about 1 mol dm^{-3}
- starch indicator
- any other required apparatus normally found in a college laboratory.

Your plan should include details of, including quantities:

- the preparation of 250.0 cm^3 of 0.075 mol dm^{-3} aqueous $Na_2S_2O_3$;
- the preparation of about 100 cm^3 of a saturated solution of calcium iodate, $Ca(IO_3)_2$ at 20 °C;
- the essential details of the titration process;
- an outline of how you would use your mean titre value to determine the solubility product of $Ca(IO_3)_2$.

In your calculations, you should let **V** cm^3 be your mean titre and express your final mathematical expression in terms of **V**.

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[8]

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(b) The solubility of calcium iodate(V), $Ca(IO_3)_2$, at 20 °C, is approximately 2.4 g dm^{-3}. Justify, with calculations, that the chosen concentration, 0.075 mol dm^{-3}, of aqueous $Na_2S_2O_3$ solution is appropriate. [M_r of $Ca(IO_3)_2$ = 389.9]

[2]

(c) The experiment described in your plan in **(a)** is repeated using 0.1 mol dm^{-3} aqueous $Ca(NO_3)_2$ solution, instead of deionised water, to prepare a saturated solution of calcium iodate(V).

State and explain how, if at all, the titre values and calculated K_{sp} would be expected to differ from that obtained in **(a)**.

[Assume that both the experiments were carried out under the same conditions.]

....

....

....

....

....

....

....

[2]

[Total: 12]

For Examiner's Use

4 Qualitative Analysis

(a) You are provided with solid **FA 6** which contains $S_2O_8^{2-}$ anion and one cation from the ions listed in the *Qualitative Analysis Notes*.

You are to perform the tests described in Table 4.1 to deduce the nature of **FA 6** and to identify the cation present in **FA 6**.

Record your observations in Table 4.1. Your answers should include
- details of colour changes and precipitates formed,
- the identity of any gases evolved, and details of the test used to identify each one.

You should indicate clearly at what stage in a test a change occurs.
Marks are not given for chemical equations.
No additional or confirmatory tests for ions present should be attempted.

Table 4.1

Tests	*Observations*
(i) To a spatula measure of **FA 6** in a boiling tube, add 1-2 cm depth of dilute hydrochloric acid and heat cautiously. Retain the solution for **test (ii)**.	
(ii) To the resulting solution from **test (i)**, add barium nitrate(V) solution.	
(iii) To a spatula measure of **FA 6** in a test tube, add 1-2 cm depth of freshly prepared solution of iron(II) sulfate solution and warm.	
(iv) Place a spatula measure of **FA 6** in a boiling tube and add dilute nitric acid to dissolve the solid. Then add 1 cm depth of manganese(II) sulfate solution and 4 drops of silver nitrate(V) solution to act as a catalyst. Heat cautiously to bring the mixture to boiling.	
(v) To a spatula measure of **FA 6** in a boiling tube, add 1 cm depth of aqueous sodium hydroxide. Heat cautiously.	

[5]

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(b) **(i)** State, with supporting evidence, the nature of **FA 6**.

nature of **FA 6**: ………………………………………………

evidence: ……………………………………………………………………..

………………………………………………………………………………

………………………………………………………………………………

(ii) Identify the cation in **FA 6**. cation: ………………

(iii) Write the ionic equation, including state symbols, for the reaction that occurs in **test (i)**.

……………………………………………………………………………… [4]

(c) You are provided with an organic solution **FA 7** which contains one functional group.

Care: FA 7 is flammable. Do not use Bunsen burner for heating. Use the hot water provided.

FA 7 gives a positive test with 2,4–dinitrophenylhydrazine.

Devise **two** other confirmatory tests using the **bench reagents provided** to identify the functional group present in **FA 7**.

Carry out the tests and record details of the tests performed and observations made in table 4.2.

Table 4.2

Confirmatory Tests	*Observations*

Functional group present in **FA 7**: …………..……………… [5]

[Total: 14]

Qualitative Analysis Notes

[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	*reaction with*	
	NaOH(aq)	NH_3(aq)
aluminium, Al^{3+}(aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH_4^+(aq)	ammonia produced on heating	–
barium, Ba^{2+}(aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca^{2+}(aq)	white ppt. with high [Ca^{2+}(aq)]	no. ppt.
chromium(III), Cr^{3+}(aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu^{2+}(aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe^{2+}(aq)	green ppt., turning brown on contact with air insoluble in excess	green ppt., turning brown on contact with air insoluble in excess
iron(III), Fe^{3+}(aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg^{2+}(aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn^{2+}(aq)	off-white ppt., rapidly turning brown on contact with air insoluble in excess	off-white ppt., rapidly turning brown on contact with air insoluble in excess
zinc, Zn^{2+}(aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

anion	*reaction*
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $Cl^-(aq)$	gives white ppt. with $Ag^+(aq)$ (soluble in $NH_3(aq)$)
bromide, $Br^-(aq)$	gives pale cream ppt. with $Ag^+(aq)$ (partially soluble in $NH_3(aq)$)
iodide, $I^-(aq)$	gives yellow ppt. with $Ag^+(aq)$ (insoluble in $NH_3(aq)$)
nitrate, $NO_3^-(aq)$	NH_3 liberated on heating with $OH^-(aq)$ and A*l* foil
nitrite, $NO_2^-(aq)$	NH_3 liberated on heating with $OH^-(aq)$ and A*l* foil; NO liberated by dilute acids (colourless NO $\rightarrow$ (pale) brown NO_2 in air)
sulfate, $SO_4^{2-}(aq)$	gives white ppt. with $Ba^{2+}(aq)$ (insoluble in excess dilute strong acids)
sulfite, $SO_3^{2-}(aq)$	SO_2 liberated with dilute acids; gives white ppt. with $Ba^{2+}(aq)$ (soluble in excess dilute strong acids)

(c) Tests for gases

gas	*test and test results*
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

halogen	*colour of element*	*colour in aqueous solution*	*colour in hexane*
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple